Challenge	What we found and did
Query fan-out	The first service-record view issued one request per aircraft: 1 + 118 requests on load. It now loads two page-initiated calls, and the assets list is one aggregation query (about 1.0 s on staging).
Cost of conflict analysis	On the real corpus, 1,835 of 2,079 records have a candidate. Classifying the top pair of each with five self-consistency votes projects to 9,175 model requests, against a free-tier limit of 1,000 per day: at least ten days of quota. An opt-in cascade (a deterministic text screen first, then two votes, escalating to five only on disagreement) projects to 2,062–5,155 requests. Token cost has not been measured on the full corpus.
Model instability	On one pair, two models gave different labels (`contradicts` versus `scope_disjoint`). We record method, model and votes for every classification and do not tune thresholds to look better.
Legacy placeholder data	Records ingested by an early version carried a confidence of 1.0 and status auto-accepted without ever being analysed. The API now reports such untouched values as unanalysed instead of displaying 1.0.
Provider limits	Rate limiting (HTTP 429) is normal: a 13-record run met 27 and retried each. Retries are bounded, a daily budget stops runs cleanly, and a missing live model fails closed.
Empty model output	On staging, fresh diagnoses returned no explanation text; the cause is not established (the staging model is smaller than the one used locally). The API treats empty model output as an explicit error (HTTP 503) and never as a blank success.
Dense graphs	The aircraft with 153 records produces a 305-edge subgraph that is unreadable if drawn with everything else; the interface shows one aircraft at a time.